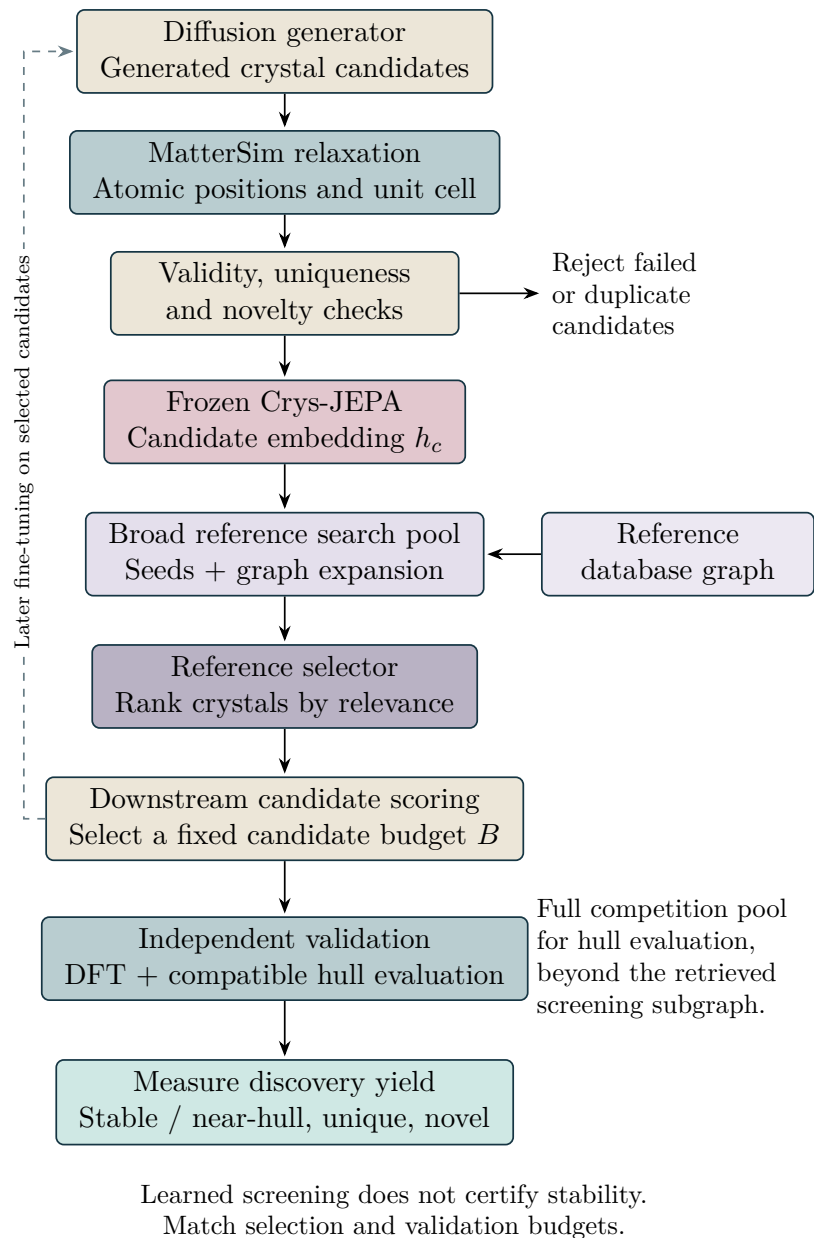
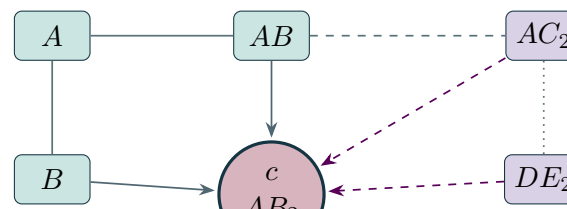


(a) Candidate screening and validation

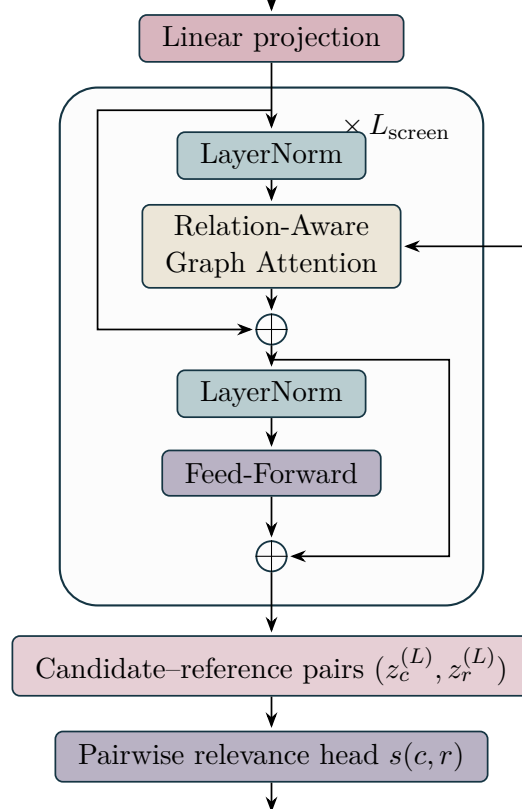


(b) Candidate-conditioned reference selector

Candidate c and whole-crystal references in the search pool



Pink: candidate query node
Teal: competition references
Purple: cross-chemistry analogues
Solid links: chemical context
Dashed links: analogue context
Dotted links: structural similarity



For candidate and reference nodes:
 $x_i = [h_i^{\text{Crys-JEPA}}; \text{composition}_i]$

E, τ_{ij}, e_{ij}
Edges, relation types and similarities

Reference nodes exchange information. The candidate receives competition and analogue context.

$s(c, r)$ measures reference relevance, not the candidate's stability. Training objective remains open. Possible supervision: usefulness for downstream screening. Selection rule and role budgets must be specified and evaluated.

Rank and select k references: $\{(r, s(c, r))\}$

Proposed reference-selection flow. Search rules, relevance supervision and the downstream candidate scorer remain design choices.